Code No: 138GK **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, June - 2024
MICROPROCESSORS AND MICROCONTROLLERS
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	Name the different segments available in 8086 µ*p*.	[2]
b)	Write any three branch and call instructions.	[3]
c)	Explain PUSH and POP instructions in 8051.	[2]
d)	Explain the memory organization of 8051.	[3]
e)	What is the need of DMA?	[2]
f)	Write a short note on SPI bus.	[3]
g)	What is exception?	[2]
h)	Write the role of barrel shifter in ARM.	[3]
i)	List the applications of DSP processors.	[2]
j)	Explain the priority concept in Cortex processor.	[3]

PART – B

(50 Marks)

2.a) What is Interrupt? Explain the Interrupts of 8086.
b) Define the macro? Explain the working of macros in 8086 with one example. [5+5]

OR

3. Name the different addressing modes of 8086 and explain them with suitable example. [10]

4.a) Explain the TCON special function register with a diagram in 8051 microcontroller.
b) Explain the serial communication interrupts. [5+5]

OR

5.a) How does 8051 differentiate between the external and internal program memory.
b) Discuss the I/O Ports of 8051. [5+5]

6.a) Draw the 8251 USART architecture and explain its operation.
b) Draw a neat sketch of DAC to be interfaced with 8051 microcontroller. [5+5]

OR

7.a) Explain about on board communication Interfaces.
b) Write short notes on serial communication standards. [6+4]

8.a) Discuss the main features of the ARM Processor.
b) Explain the concept of thumb instructions in ARM processor. [5+5]

OR

9. List the various modes of ARM processor and explain any two modes of it. [10]

10. Draw the functional block diagram of OMAP Processor and explain working of all blocks available in it. [10]

OR

11.a) Explain the main features of the OMAP Processor.
b) Discuss about memory access attributes of OMAP processor. [5+5]

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